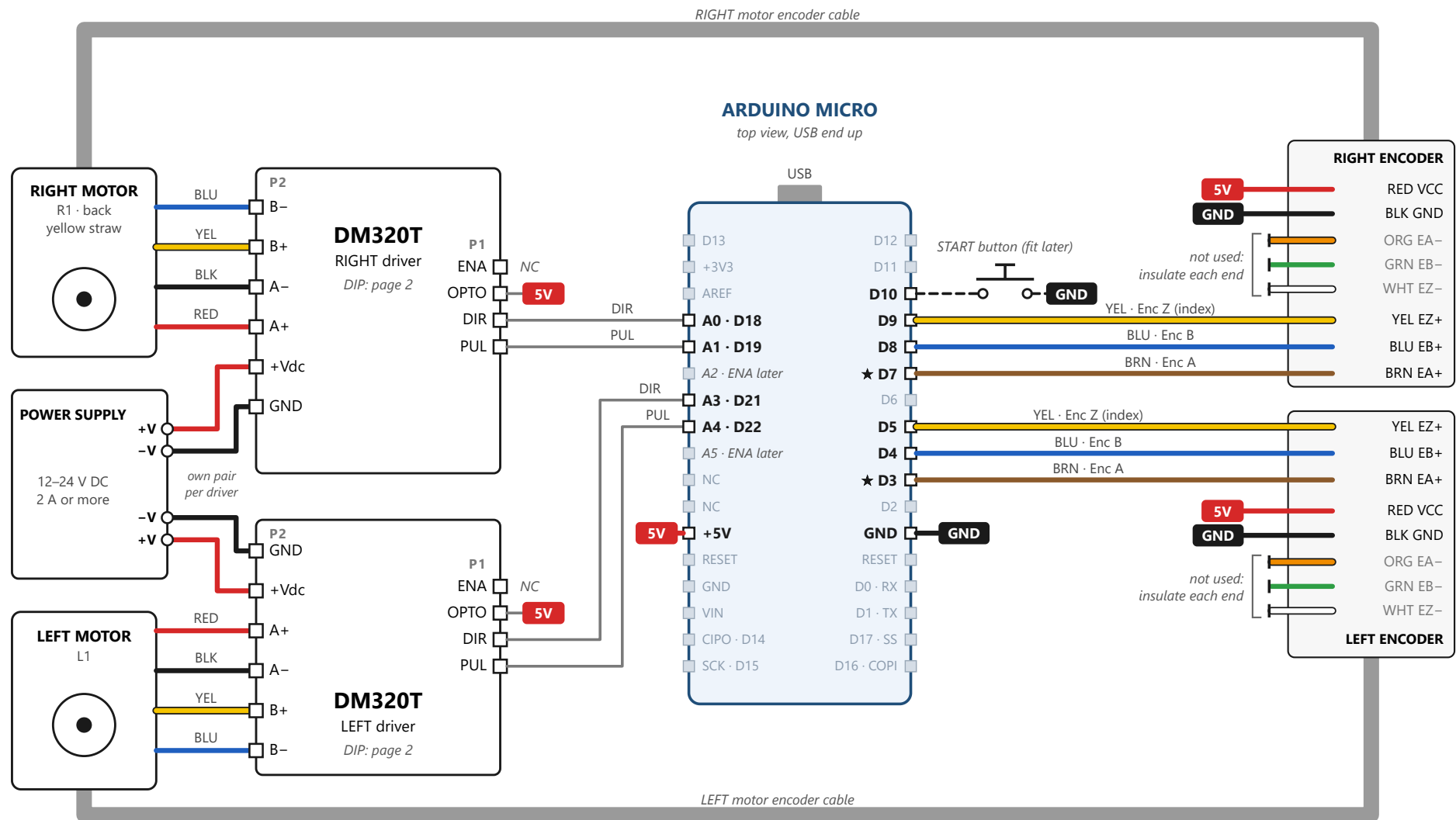




Tags: every **5V** joins together, and every **GND** joins together. The Micro has one +5V pin (left-hand header), so split it 4 ways (2 × OPTO, 2 × encoder RED) with a terminal strip or lever connectors. The encoder BLK wires go to the GND pin on the right-hand header (either GND works). The PSU –V does **not** connect to the Micro.

⚠ Never plug or unplug the green terminal blocks while powered.

Keep PUL/DIR and encoder wires ≥ 10 cm from the motor leads. The motor and encoder cables both use RED/BLK/YEL/BLU, so label them.

Connection checklist and driver settings

Tick each connection as you make it

LEFT motor (L1)

MOTOR LEADS → LEFT DRIVER P2

- ☐  RED → A+
- ☐  BLK → A-
- ☐  YEL → B+
- ☐  BLU → B-

MICRO → LEFT DRIVER P1

- ☐ A4 · D22 → PUL
- ☐ A3 · D21 → DIR
- ☐ 5V → OPTO
- ☐ nothing → ENA

ENCODER CABLE → MICRO

- ☐  RED · VCC → 5V
- ☐  BLK · GND → GND
- ☐  BRN · EA+ → D3 ★
- ☐  BLU · EB+ → D4
- ☐  YEL · EZ+ → D5
- ☐ ORG, GRN, WHT → insulate, not connected

Pins in main.cpp

```
AccelStepper stepperLeft (DRIVER, A4, A3); // PUL, DIR
AccelStepper stepperRight(DRIVER, A1, A0); // PUL, DIR
Encoder encoderLeft (3, 4); // index Z: D5
Encoder encoderRight(7, 8); // index Z: D9
// Reserved: ENA left A5, right A2;
// START button D10 (not wired yet)
```

RIGHT motor (R1, back, yellow straw)

MOTOR LEADS → RIGHT DRIVER P2

- ☐  RED → A+
- ☐  BLK → A-
- ☐  YEL → B+
- ☐  BLU → B-

MICRO → RIGHT DRIVER P1

- ☐ A1 · D19 → PUL
- ☐ A0 · D18 → DIR
- ☐ 5V → OPTO
- ☐ nothing → ENA

ENCODER CABLE → MICRO

- ☐  RED · VCC → 5V
- ☐  BLK · GND → GND
- ☐  BRN · EA+ → D7 ★
- ☐  BLU · EB+ → D8
- ☐  YEL · EZ+ → D9
- ☐ ORG, GRN, WHT → insulate, not connected

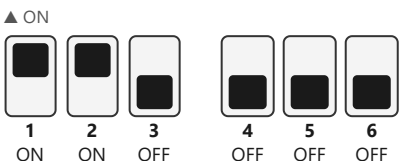
Notes

- **No resistors needed.** The DM320T inputs are opto-isolated and wired common-anode: OPTO goes to +5 V, and the Micro switches PUL and DIR by pulling them low (manual fig. 2: R = 0 at 5 V).
- **Grounds.** The driver power ground stays separate from the Arduino; the opto inputs isolate them.
- **Supply.** Run a separate pair of wires from the PSU to each driver (no daisy-chaining). Lower voltage (12 V) is quieter at slow speeds. Fit the PSU's mains cover.
- **Encoder supply.** 3.3–5.5 V, so the Micro's 5 V is fine. Outputs are 5 V logic and connect directly to the Micro (StepperOnline listing).

Power

- ☐ PSU +V → Left driver +Vdc
- ☐ PSU -V → Left driver GND
- ☐ PSU +V → Right driver +Vdc
- ☐ PSU -V → Right driver GND
- ☐ Micro USB → PC or USB charger
- ☐ D10 → START button → GND (fit later)

DIP switches (both drivers the same)



SW1–3

Current 1.3 A peak (0.92 A RMS)
Motor is rated 1.0 A; the manual says × 1.4

SW4–6

8000 steps/rev (40 microsteps)
Exactly 2 steps per encoder count

Change switches only with power off.

- **Unused encoder wires** (ORG, GRN, WHT) are driven outputs, so insulate each one separately. Don't ground them.
- **Driver red LED:** 1 flash = over-current, 2 flashes = over-voltage (> 34 V). Turn the power off and on again to reset.
- **⚠ Start-up order:** line up both straws in the start pose with everything off, plug in the Micro (0° is set here), switch on the 12 V, then press r. Any jerk as the drivers power up is corrected. Motor and encoder directions checked 26 Sep 2026.